

# Lattice — Feature Showcase

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Open this file with **Lattice** to see every feature rendered live.

Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.

Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

## Table of Contents

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# Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

| Style         | Syntax                                  | Result                   |
|---------------|-----------------------------------------|--------------------------|
| Bold          | <code>**bold**</code>                   | <b>bold</b>              |
| Italic        | <code>*italic*</code>                   | <i>italic</i>            |
| Bold + italic | <code>***both***</code>                 | <b><i>both</i></b>       |
| Strikethrough | <code>~~strikethrough~~</code>          | <del>strikethrough</del> |
| Inline code   | <code>`code`</code>                     | code                     |
| Superscript   | <code>X&lt;sup&gt;2&lt;/sup&gt;</code>  | X <sup>2</sup>           |
| Subscript     | <code>H&lt;sub&gt;2&lt;/sub&gt;O</code> | H <sub>2</sub> O         |

**Note — sub/superscript:** CommonMark/GFM has no native sub/superscript syntax. Lattice renders `<sup>` and `<sub>` HTML tags directly (GitHub and Obsidian do too). We recommend, though that you use alternatives that work everywhere: KaTeX math ( `$X^2$` →  $X^2$ , `$H_{20}$` →  $H_2O$ ) or Unicode characters ( `x2` , `H2O` ). The linter flags these tags as a portability hint — they may not render in all renderers.

# Highlight Marks

**Highlight marks** are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a **bright amber background**.

This works in **any context** — inside a sentence, `==across italic and bold spans==`, or on its own line:

**This entire sentence is highlighted.**

[!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

# Lists

## Unordered

- First item
- Second item
  - Nested item A
  - Nested item B
    - Deeply nested
- Third item with **bold** and *italic* text

## Ordered

1. Install Lattice from the [releases page](#)
2. Open any `.md` file — double-click or via **File → Open**
3. Choose your view:
  - i. **Edit** — raw CodeMirror editor only
  - ii. **Preview** — rendered HTML only
  - iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

## Definition-style (using bold + indent)

**Local-first** All files stay on your machine. No cloud sync required.

**Documents as Code** Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

## Task Lists (GFM)

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GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- ☒ **Install Lattice**
- ☒ **Open the demo file**
- ☒ **Explore Edit / Preview / Dual views**
- ☐ Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- ☐ Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- ☐ Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- ☒ **Open a second file in a new window (multi-window support)**
- ☐ Try printing to PDF with `Ctrl/Cmd+P`

# Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

| Feature               | Edit View | Preview View | Dual View |
|-----------------------|-----------|--------------|-----------|
| CodeMirror editor     | ✓         | —            | ✓         |
| Live rendered preview | —         | ✓            | ✓         |
| Scroll sync           | —         | —            | ✓         |
| Print (Ctrl+P)        | ✓         | ✓            | ✓         |
| Highlight marks       | ✓         | ✓            | ✓         |
| Mermaid diagrams      | —         | ✓            | ✓         |
| KaTeX mathematics     | —         | ✓            | ✓         |

# Code

## Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

## Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin - converts ==text== to <mark> elements
export const MARK_RE = /==(?:[^\n]|(?!=))+?)/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
      parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
    parts.push({
```

```

        type: 'element', tagName: 'mark',
        properties: {},
        children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
}
if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
return parts;
}

// Tauri command — read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
    let content = fs::read_to_string(&path)
        .map_err(|e| format!("Cannot read {path}: {e}"))?;
    let hash = compute_sha256(&content);
    Ok(FileReadResult { content, hash })
}

# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build

```

## Blockquotes

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**Single-level blockquote.** Great for callouts, citations, or side notes.

Nested blockquotes are also supported:

Inner quote — *"Documents as code, not documents as data."*

— Lattice design philosophy

## GitHub-style Alerts (rendered by GitHub and Lattice)

[!NOTE] Useful information that users should know, even when skimming.

[!TIP] Optional advice for a better experience.

[!IMPORTANT] Key information users need to know to achieve their goal.

[!WARNING] Urgent information that needs immediate attention.

[!CAUTION] Advises about risks or negative outcomes.

## Mathematics — KaTeX

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Lattice renders mathematics via **KaTeX**. Use  $\dots$  for inline and  $\displaystyle\dots$  for display math.

### Inline math

Einstein's famous equation  $E = mc^2$  and the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  render inline.

The Euler identity  $e^{i\pi} + 1 = 0$  is often called the most beautiful equation in mathematics.

### Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$
$$\frac{d}{dx} \left[ \int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

### Display math — Linear Algebra

$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

### Display math — Series & Limits

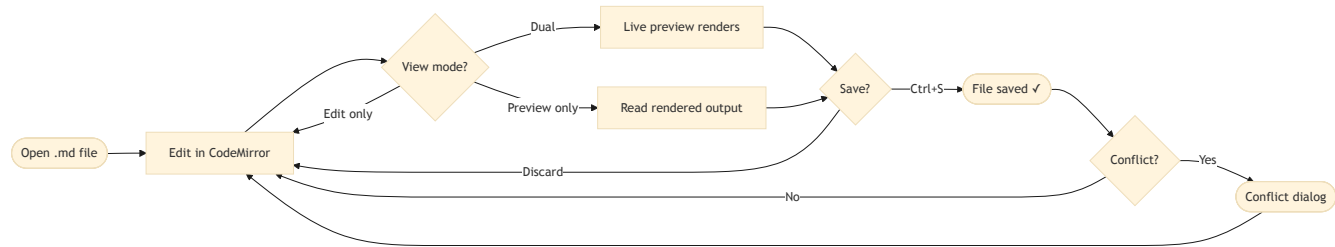
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$
$$\lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e$$

## Mermaid Diagrams

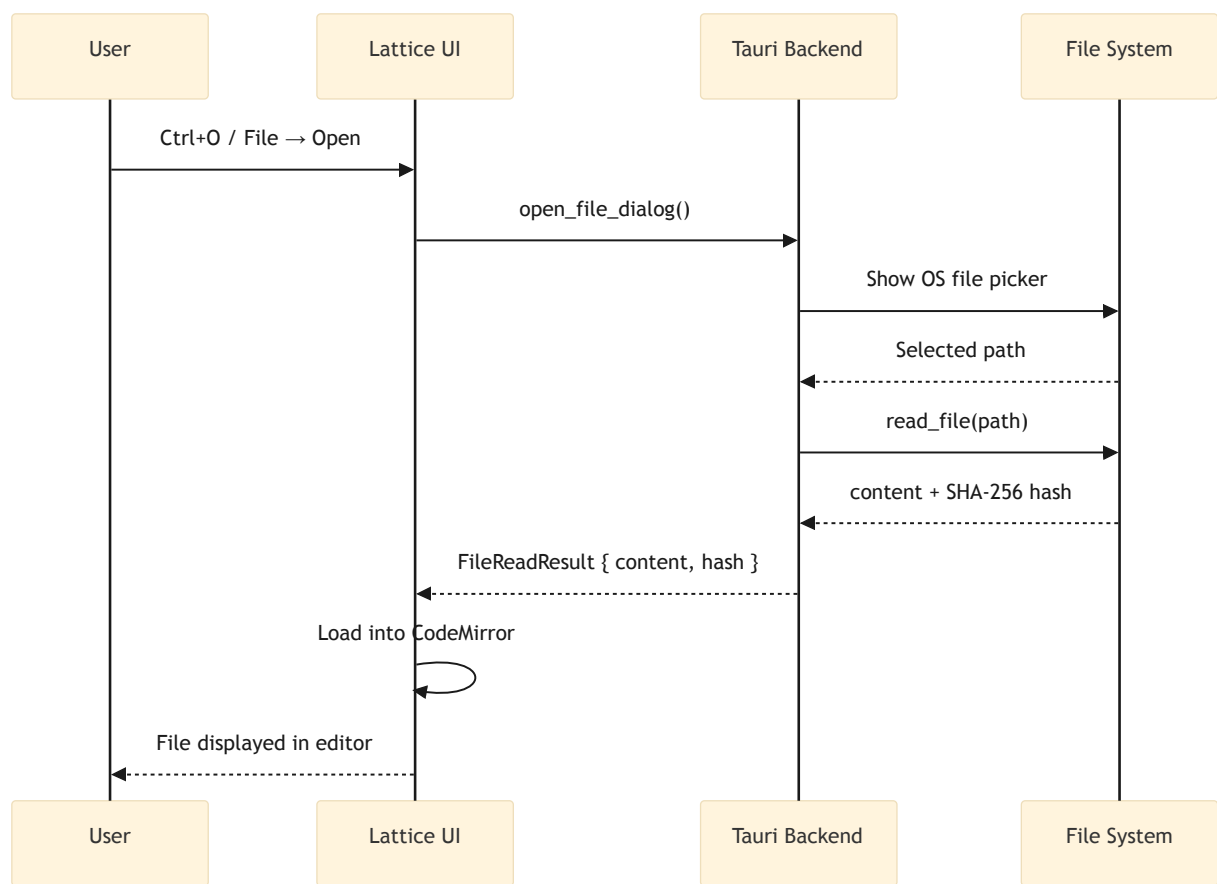
---

Lattice renders ````mermaid```` fenced blocks inline in the preview.

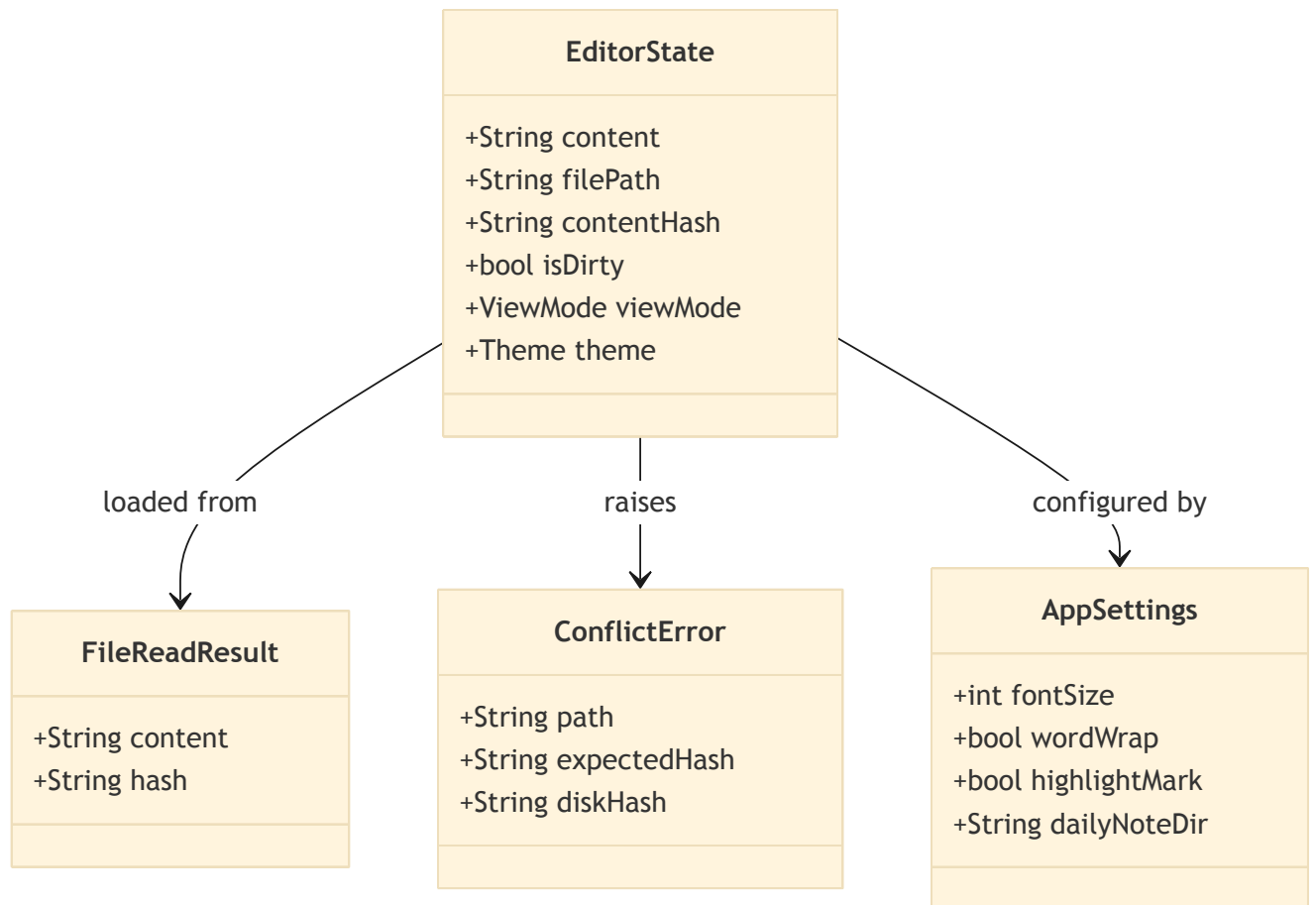
## Flowchart — Lattice Edit Cycle



## Sequence Diagram — File Open Flow

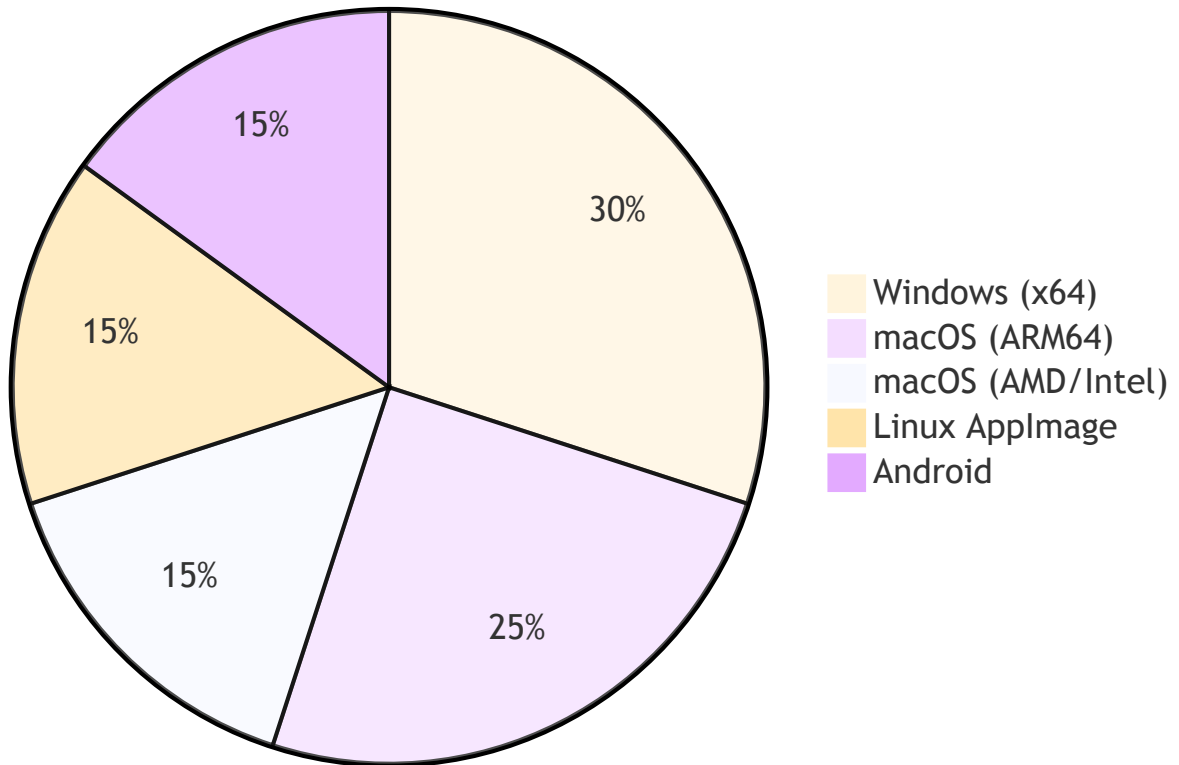


## Class Diagram — Core Data Model



## Pie Chart — Supported Platforms

## Lattice Platform Coverage



## Images

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Paste images directly into Lattice with `Ctrl/Cmd+V`.

The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
![[Image]](demo_assets/img_1780574419628.png)
```

```
![[Image]](demo_assets/img_1780575215039.png)
```

Here is how it renders in Dual (edit on the left) mode:

Image

Here is how it renders in Dual (edit on the bottom) mode:



**Tip:** The assets folder name is derived from the document filename, keeping related files grouped together and easy to version-control.

## Links

- [Lattice on GitHub](#) — source code and issue tracker
- [Releases Page](#) — download installers
- [Release Site](#) — GitHub Pages release portal
- Relative link: [System Architecture Demo](#)
- Relative link: [Physics Equations Demo](#)

Auto-link: <https://github.com/blessia-blessini/lattice>

## HTML in Markdown

Lattice renders a **safe whitelist** of inline HTML tags and passes everything else through as literal text. The linter flags all inline HTML — whitelisted tags get a **hint** (portability risk), non-whitelisted tags get a **warning** (not rendered here).

| HTML element | Lattice preview | GitHub | Portable alternative                                   |
|--------------|-----------------|--------|--------------------------------------------------------|
| <sup>        | ✓ rendered      | ✓      | KaTeX: $\$X^2\$ \rightarrow X^2$ · Unicode: $x^2$      |
| <sub>        | ✓ rendered      | ✓      | KaTeX: $\$H_{20}\$ \rightarrow H_2O$ · Unicode: $H_2O$ |
| <kbd>        | ✓ rendered      | ✓      | Inline code: <code>`Ctrl+S`</code>                     |
| <br>         | ✓ rendered      | ✓      | Two trailing spaces or blank line                      |
| <mark>       | ⚠ plain text    | ✓      | <code>==highlight==</code> (Lattice-native)            |
| <details>    | ⚠ plain text    | ✓      | Headings + TOC (no universal equivalent)               |
| Other HTML   | ⚠ plain text    | varies | Markdown equivalent (case-by-case)                     |

Live examples of the whitelisted tags:

- Superscript:  $X^2 + Y^3$
- Subscript:  $H_2O$  and  $CO_2$

- Keyboard: press `Ctrl+S` to save

## Keyboard Shortcuts Reference

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| Action                     | Windows / Linux           | macOS                    |
|----------------------------|---------------------------|--------------------------|
| Save                       | <code>Ctrl+S</code>       | <code>Cmd+S</code>       |
| Open file                  | <code>Ctrl+O</code>       | <code>Cmd+O</code>       |
| New window                 | <code>Ctrl+N</code>       | <code>Cmd+N</code>       |
| Generate / refresh TOC     | <code>Ctrl+Shift+T</code> | <code>Cmd+Shift+T</code> |
| Auto-align table           | <code>Ctrl+Shift+L</code> | <code>Cmd+Shift+L</code> |
| Paste image from clipboard | <code>Ctrl+V</code>       | <code>Cmd+V</code>       |
| Print / Save as PDF        | <code>Ctrl+P</code>       | <code>Cmd+P</code>       |
| Increase font size         | <code>Ctrl++</code>       | <code>Cmd++</code>       |
| Decrease font size         | <code>Ctrl+-</code>       | <code>Cmd+-</code>       |